

SYSTEM ARCHITECTURE OVERVIEW — FULL-BODY CONTEXTUAL PHYSIOLOGY NODE (NODE 5)

OVERVIEW OF THE DISCLOSED SYSTEM

The present disclosure introduces a full-body contextual physiology monitoring system configured as a distributed sensing node within a multi-node intimate physiology platform. The disclosed system, referred to herein as Node 5, operates as a systemic reference layer that captures whole-body physiological signals used to interpret localized anatomical measurements obtained from companion sensing devices. Node 5 may function independently or as part of an integrated physiological ecosystem comprising multiple coordinated sensing nodes. Each node may collect different categories of physiological information, while Node 5 provides contextual reference data enabling cross-correlation and interpretation of intimate physiological states. WEARABLE IMPLEMENTATION ARCHITECTURES

In various embodiments, Node 5 may be implemented in one or more wearable configurations, including but not limited to:

- a finger-worn ring configured for continuous physiological monitoring;
- a chest-mounted patch positioned over the cardiac or thoracic region;
- a wrist-mounted wearable device;
- a textile-integrated biosensing module;
- a pendant, necklace, or body-mounted module;
- an adhesive conformal patch;
- or any body-coupled sensing platform capable of acquiring systemic physiological data. These implementations may share common sensing, computation, and communication architectures while differing in placement, ergonomics, power delivery mechanisms, and sensing priorities. SENSOR ARCHITECTURE

Node 5 may incorporate one or more physiological sensing subsystems, including but not limited to photoplethysmography (PPG), electrocardiography (ECG), bioimpedance sensing, skin temperature monitoring, electrodermal activity sensing, respiratory motion detection, accelerometry, gyroscopic sensing, environmental sensing, pressure sensing, or other physiological acquisition modalities. Sensor data may be sampled continuously, periodically, event-triggered, or adaptively based on detected physiological conditions or system power optimization strategies. DATA PROCESSING AND SYNCHRONIZATION

Collected sensor data may be processed locally on-device, partially processed at an edge processor, or transmitted to remote computational infrastructure. Node 5 may synchronize temporally with additional physiological nodes to enable correlated interpretation of distributed physiological measurements. Synchronization may occur via wireless communication protocols, time-stamped event alignment, shared physiological markers, predictive synchronization algorithms, or machine learning-based temporal alignment models. CONTEXTUAL PHYSIOLOGY LAYER

Node 5 establishes a contextual physiology layer configured to generate systemic reference metrics describing autonomic balance, cardiovascular readiness, emotional engagement, recovery state, and environmental interaction context. These contextual outputs enable interpretation of intimate physiology measurements that would otherwise lack sufficient system-level meaning. The disclosed architecture therefore transforms conventional wearable sensing from isolated signal acquisition into coordinated physiological interpretation across multiple anatomical and behavioral domains.